

Layout Design Basis Report — General Arrangement Notes

Project: Hyperscale DC — Chennai, 48 MW, Tier III (Concurrently Maintainable, N+1)

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SAMPLE / SYNTHETIC DATA — representative of a Chennai hyperscale data-centre general arrangement, generated for the SiteMind demo. Room and equipment figures are fictional but internally consistent; standards citations are real (see standards/spatial_clauses.json).

General Notes

Note 1: This Layout Design Basis Report governs the general arrangement of the "Hyperscale DC — Chennai, 48 MW, Tier III (Concurrently Maintainable, N+1)" project and shall be read with the latest revision of the architectural floor plans.

Note 2: Electrical clearances noted below conform to Regulation 37 of the Central Electricity Authority (Measures Relating to Safety and Electric Supply) Regulations, 2010; means of egress noted below conform to Part 4 of the National Building Code of India, 2016.

Note 3: Data Hall 1 measures 30 m by 20 m in plan and houses the primary IT rack rows. Rack-row and cold-aisle/hot-aisle geometry inside Data Hall 1 is governed by ASHRAE TC 9.9, which is not a freely redistributable standard and is not digitised in this platform's standards library, and is shown on the floor map for spatial context only — not for a compliance verdict.

Note 4: The LV Switchroom measures 12 m x 8 m in plan and contains the low-voltage distribution switchboard serving Data Hall 1.

Note 5: A front clearance of 0.8 m is provided for the LV distribution panel, allowing operator access for breaker racking and de-racking.

Note 6: A rear clearance of 0.9 m is maintained behind the LV panel row, and the passage behind the panel remains clear to a height of 2.1 m for maintenance access.

Note 7: The Cooling Plant Room measures 14 m x 10 m in plan and houses the chilled-water pump skids and CRAC distribution headers serving Data Hall 1.

Note 8: The corridor immediately outside the LV Switchroom terminates in a dead end; the dead-end run measures 18 m from the last point of access to an alternative exit route, to the terminating wall.

Note 9: An exit door in the corridor is 1200 mm wide, located in the north wall.

Note 10: Occupant-load figures for the corridor and Data Hall 1, and the travel distance from the LV Switchroom to the nearest final exit, are being finalised with the fire consultant and are not included in this revision; the required exit width and travel-distance checks cannot be completed until those figures are issued.

Note 11: The LV Switchroom sits immediately to the west of Data Hall 1, sharing a common wall along Data Hall 1's western elevation.

Note 12: The Cooling Plant Room is located on the east side of Data Hall 1, adjoining it directly to keep the chilled-water pipe run to the CRAC headers as short as practicable.

Note 13: The corridor runs along the north face of the LV Switchroom, giving operations staff direct access to the switchroom's front clearance zone.